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Implicit Cognitive Processes as Aptitudes for Learning

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Following the research traditions of Richard Snow, several constructs from cognitive theory and research are considered with respect to their potential as meaningful aptitudes for learning. Recent research has suggested that implicit learning and memory processes are relatively distinct in character from explicit, declarative memory processes that have been the primary focus of learning theories for most of the past century. Two forms of implicit cognitive processes are considered for their utility as aptitude constructs that help to explain learner differences in complex forms of learning. Evidence of individual differences in repetition priming suggests that these implicit processes may represent basic operations underlying cognitive skill acquisition. Evidence of individual differences in semantic priming suggests that these implicit processes may represent important cognitive constraints in complex language tasks. These ideas are considered in the context of Richard Snow's views on aptitude theory and the importance of advancing our understanding of learner differences in aptitude processes.

Richard Snow spent much of his distinguished career shaping the way psychologists and educators think about aptitudes for learning. The term *aptitude* has often polarized subsets of theorists and practitioners. However, Snow's view of aptitudes for learning was broad and inclusive of many contextual issues. He articulated the need for instructional theory to understand learners' readiness for learning in different contexts by emphasizing the importance of person-by-situation interactions (Snow, 1992, 1994). In addition, he promoted a broad definition of aptitude that included cognitive, affective, and conative processes (Snow, 1992; Snow, Corno, & Jackson, 1996; Snow & Jackson, 1994).

In acknowledging contrasting viewpoints toward aptitude research, Snow seemed to take a pragmatic stance as exemplified in his early writing, "Educational researchers not interested in this problem can ignore it. But individual differences in aptitude for learning will not go away" (Snow, 1977, p. 99). Rather than ignore it, he chose to pursue the understanding of learner differences with the goal of improving educators' ability to design and adapt instruction in light of these differences. His many contributions in these areas are well chronicled in a recent volume by former students and colleagues (Corno et al., 2002).

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Two of the many principles of aptitude research promoted by Richard Snow provide the underlying motivation for this article. The first principle is that aptitudes for learning should be investigated in terms of processes as opposed to traits or absolute performance levels (Snow, 1980, 1991, 1992). In this way, research can contribute to an understanding of how aptitude constructs map on to learning processes in different instructional contexts. Second, aptitude research should seek to refine, expand, and reconceptualize existing aptitude constructs in light of new psychological and educational theory (Snow, 1977, 1980). Snow was not particularly optimistic that new aptitudes would be discovered through such efforts. He did not rule out this possibility, and I believe he viewed a search for new aptitudes as a worthwhile endeavor. However, he most often described how old aptitude concepts should and will be reconceptualized continually. In his early writing on aptitude research, Snow (1977) summarized this view in saying, "We can strive for, and expect, new aptitude constructs, but these are likely to be woven in large part from threads of existing cloth" (p. 53).

Along with a realistic view of whether truly new aptitude constructs were likely to be found, Snow's writings reflected an underlying desire to identify areas of aptitude research that might change conventional understanding of learner differences. This is best exemplified in his pioneering work in the domains of conative and affective processes, which lie outside of the long-standing traditions of cognitive aptitude research (Snow, 1989, 1991, 1992; Snow et al., 1996; Snow & Jackson, 1994). Although he devoted much of his later efforts to aptitude theory in the conative and affect domains, he also continued to theorize about cognitive aptitudes (Snow, 1998; Snow & Lohman, 1989, 1993). This continued interest in refining the understanding of cognitive aptitude processes exemplified his commitment to the principle that aptitude theories, no matter how well developed, continually need to be reconceptualized in light of new evidence and new psychological theory.

In this article I consider the possibility of new cognitive aptitudes based on recent theory and evidence from memory research. The term *new* is used with some reluctance, because as Snow eloquently stated, it seems unlikely that there are any psychological concepts that have not been investigated previously in one form or another. I use the term new only in the sense that these constructs appear to represent something more than a rethinking of conventional aptitudes that are central in reviews of cognitive ability research such as Carroll (1993).

The domain of aptitude constructs discussed here is referred to as *implicit cognitive processes.* I use this term to refer to a set of learning and memory processes that operate primarily outside of the realm of awareness. Memory processes underlying most conventional measures of recall and recognition are commonly described as *explicit* because retrieval typically includes (or perhaps depends on) effortful search and some level of awareness of the prior learning events. In contrast, implicit memory processes are revealed in performance facilitation, often without content-specific retrieval intent by the learner and despite lack of awareness of the original learning event or events. For example, native language speakers typically are unaware of the grammatical rules that they follow. When asked, many speakers cannot accurately describe these rules, and they are almost certainly unaware of the learning events in which they were acquired. Furthermore, they typically have no conscious intent to speak grammatically during most speech acts. Yet, a proficient speaker obeys such rules flawlessly without attention to or explicit knowledge of this facet of speech. As such, the memory processes underlying such performance could be described as implicit.

Implicit cognitive processes are thought to operate in concert with explicit processes within most complex learning and performance activities. However, growing evidence suggests that these two broad categories of learning and memory processes have some degree of functional and even structural independence. The theory and evidence for the distinctive nature of implicit processes come primarily from cognitive neuroscience research. Within this area of research, phenomena that I discuss here have been termed *implicit memory* (Kirsner, 1998; Schacter, 1992), *implicit learning* (Reber, 1989; Seger, 1994; Stadler & Frensch, 1998), *procedural knowledge* (Anderson, 1983, 1993; Willingham, Nissen, & Bullemer, 1989), and *priming* (McNamara, 1992; Tulving & Schacter, 1990). Although the concepts represented by these various terms are distinct in certain respects, in this article I am considering the general processing domain that they represent as a whole.

There are several reasons for suggesting that the general domain of implicit cognitive processes may be a fruitful area in which to investigate potentially new aptitude constructs. First, most previously studied cognitive aptitudes correspond more closely to the domain of explicit, attention-driven memory processes. There can be no doubt that performance on complex cognitive aptitude measures, for example, general fluid reasoning ability measures, taps both explicit and implicit cognitive processes (e.g., attention-driven working memory processes and implicit procedural reasoning skills). However, I surmise that they are weighted heavily in favor of explicit processes. In support of this, there exists consistent evidence that conventional reasoning and general intellectual ability measures are highly correlated with attention-driven working memory measures (Engle, Tuholski, Laughlin, & Conway, 1999; Kyllonen, 1996; Kyllonen & Christal, 1990). In contrast, measures of general intellectual ability tend to have low correlations with procedural skill performance in consistent skills beyond the early stages (Ackerman, 1987, 1988), and they have low correlations with implicit memory measures of priming (Woltz, 1990b, 1999). Therefore, exploring individual differences in implicit cognitive processes could result in aptitude constructs that have minimal overlap with existing ones.

A second reason to explore implicit learning processes as a source of new aptitude constructs is that their importance may have been previously overlooked due to the fundamental nature of these processes. A hallmark of these processes is their inaccessibility to conscious awareness. Consequently, accurate introspective understanding has surely been lacking with respect to the role of implicit processes in complex learning.[1] This lack of understanding from self-reflection must hold both for learners who are describing their learning processes to educators and researchers and for theorists who are pondering the organization and interplay of various learning mechanisms. Although the past century has been filled with theoretical advances in understanding individual differences in learning processes relevant to education, it is an intriguing possibility that a potentially important and distinct class of cognitive processes has been largely ignored because of its relative inaccessibility to awareness.

The remainder of this article summarizes recent research on individual differences in two areas of implicit cognitive processes. The first pertains to implicit memory processes assessed through repetition priming. The second pertains to

[1]Given that implicit and explicit processes typically function in concert within complex activities, a strong assumption of complete inaccessibility to implicit memory processes is probably overstated (i.e., at a minimum we must have awareness of products of implicit processes when attentional processes are simultaneously engaged). However, there is sufficient evidence of dissociations between implicit memory processes and explicit awareness to support a general claim that learners have minimal conscious access to them.

processes generally described as automatic spreading activation in long-term memory. Each is considered with respect to (a) evidence for reliable individual differences and (b) theory and evidence regarding their relations to more complex forms of learning. Finally, the evidence regarding these processes is considered with respect to Snow's (1992) framework for the necessary constituents of aptitude theory.

DISTINCTIONS AMONG IMPLICIT LEARNING AND MEMORY PROCESSES

In the current memory literature, there is relatively widespread acceptance of a general distinction between forms of memory and knowledge that are classified as declarative and nondeclarative (Anderson, 1993; Baddeley, 1998; Schacter, Wagner, & Buckner, 2000; Squire, 1994; Tulving, 1999). However, there is considerable disagreement regarding the organization of different memory systems or processes both within and in addition to this general distinction. A review of these issues is beyond the scope of this article. However, a review of concepts related to the measurement of selected implicit cognitive processes is important.

Much of the empirical evidence for nondeclarative forms of memory comes from priming studies. The term *priming* refers to performance facilitation that can be attributed to a specific prior processing event. Facilitation can be measured in terms of decreased response latency or error rate, or it can be reflected in an increased probability of a particular response. Most forms of priming are thought to represent implicit memory processes in that the facilitation does not depend on recall or recognition of the prior processing event (i.e., explicit memory). Priming usually refers to facilitation from a single prior processing event. Procedural skill acquisition, which is assessed through similar measures of performance change, usually refers to facilitation attributable to multiple, related prior events.

There are several forms of priming that correspond to different methods of assessment and theoretically different cognitive mechanisms. Most priming research pertaining to implicit memory has used repetition priming methods (sometimes referred to as identity or direct priming). In repetition priming, performance on a target processing event is assessed with respect to the amount of facilitation attributed to a prior processing event that is nearly identical to the target performance event in critical features. Within this category of repetition priming, there are two major subtypes. In perceptual priming, facilitation depends on the repetition of specific perceptual operations in the prime and target events (e.g., encoding the same word presented in the same font during both events). In conceptual repetition priming, facilitation depends on the repetition of specific conceptual or semantic operations in the prime and target events (e.g., retrieving and interpreting the meaning of the same word, even if the word is presented in different fonts or different modalities). Both are described as identity or repetition priming because they involve the nearly identical repetition of some specific processes, either perceptual or conceptual or in some cases both.

Most implicit memory research has used perceptual repetition priming tasks. Consequently, theory regarding implicit memory processes has focused more on perceptual than conceptual memory representation issues (e.g., Tulving & Schacter, 1990). However, investigations of conceptual priming have demonstrated implicit memory for semantically complex cognitive activities (e.g., Blaxton, 1989). The prior emphasis on perceptual rather than conceptual priming processes is unfortunate in some respects for those interested in the potential applications of implicit memory processes to learning in educational settings. Although implicit perceptual processes may be of some relevance, implicit conceptual processes seem more likely to play important roles in a wider variety of complex learning behaviors. For this reason, I focus on conceptual repetition priming as an aptitude process.

A final category of priming has been termed *semantic or associative priming.*[2] Although this form of priming shares some attributes with conceptual repetition priming, it refers to similarity-based priming rather than identity priming, and it is generally thought to reflect a temporary rather than persistent change in memory (e.g., Meyer & Schvaneveldt, 1976). Semantic or associative priming generally refers to facilitation from a prior processing event that resembles the target event in semantic or conceptual features, but it does not contain identical semantic content and does not require a repetition of exactly the same semantic processes. Temporary facilitation observed in such priming tasks is commonly attributed to spreading activation in associative memory structures (Anderson, 1983; McNamara, 1992; Neely & Kahan, 2001). Alternative explanations have also been proposed (Dosher & Rosedale, 1989; Masson, 1995; Plaut & Booth, 2000; Ratcliff & Mckoon, 1988). In addition, some evidence suggests that semantic priming may not always be as short-lived as previously thought (Becker, Moscovitch, Behrmann, & Joordens, 1997; Joordens & Becker, 1997). Regardless of the exact nature of the underlying mechanisms and the persistence of the effects, semantic priming appears to represent implicit cognitive processes that differ in character from those underlying repetition priming in that they depend on associations in memory.

REPETITION PRIMING AND SKILL LEARNING

Research on aptitudes for skill acquisition has a long history (for a review, see Ackerman, 1987). In general, this research has suggested that although individual differences in early

[2]Some researchers make a distinction between these terms, but in this article these terms are used interchangeably.

stages of skill acquisition are related to general cognitive abilities (e.g., general intellectual abilities, working memory), performance at later stages is usually less related to them (Ackerman, 1987, 1988, 1990, 1992; Fleishman, 1972; Woltz, 1988). Evidence regarding aptitude relations in later stages of skill acquisition has been more varied. Some researchers have concluded that in many relatively simple skills, individual differences are reduced over practice, and differences that do exist reflect task-specific aptitudes (Ackerman, 1987; Fleishman, 1972). In an attempt to explain these varied relations, Ackerman (1988, 1990, 1992) proposed and supported a theory of changing cognitive aptitude relations over practice, with perceptual speed and psychomotor abilities explaining differences in middle and later stages of skill acquisition, depending on the nature of the skill and the structure of practice.

For the most part, previous research on skill-learning differences has investigated conventional aptitude constructs from the psychometric tradition. If one considers the possibility of new aptitude constructs from recent cognitive theory that might relate to skill acquisition, repetition priming seems to be an obvious choice. Several lines of evidence suggest that repetition priming and procedural skill learning have overlapping characteristics (Logan, 1990; Poldrack & Gabrieli, 2001; Poldrack, Selco, Field, & Cohen, 1999). Furthermore, they have been closely linked within neuropsychological theory (e.g., Squire, 1994).[3] The primary difference between them may be methodological; assessment of procedural skill performance represents facilitation from many prior practice events, and assessment of repetition priming generally represents facilitation from a single prior event. It is therefore possible that repetition priming represents the elemental processes of procedural memory strengthening that underlie longer term skill acquisition (Poldrack & Gabrieli, 2001). If so, and if there are stable individual differences in repetition priming, then it might represent a measurable aptitude process that could help to understand individual differences in more complex forms of skill acquisition.

Individual Differences in Repetition Priming

Relatively little evidence exists regarding individual differences in implicit processes within adults in the normal range of memory functioning. It might be assumed that individuals differ in priming, just as they differ on most cognitive performance measures. However, some have suggested that implicit forms of learning reflect primitive cognitive systems in which there are minimal individual differences (e.g., Reber, 1989). Furthermore, assessing individual differences in performance facilitation from a single prior event presents serious measurement challenges. Single-event priming effects are generally small (e.g., often less than 50 msec), and priming is typically represented in change scores that are almost always less reliable than overall performance measures.

Some of the empirical evidence regarding individual differences in priming measures is consistent with the expectation that they may not exist or be measurable. Meier and Perrig (2000) directly investigated the reliability of implicit measures (primed picture clarification and primed word-stem completion), along with common explicit memory measures (recall and recognition). In three experiments that investigated dissociations among these measures, both test–retest and alternative forms of reliability estimates were lower for the priming measures compared to the explicit memory measures. The authors suggested that this lower reliability could account for the observed dissociations in this and other research. For example, amnesic patients who differ from healthy participants on explicit memory measures may not differ reliably from healthy participants on implicit measures simply because of greater measurement error.

Buchner and Wippich (2000) also investigated the reliability (internal consistency) of several common implicit memory measures. They hypothesized that most implicit measures are less reliable than explicit measures primarily because their instructions place fewer constraints on performance processes. For example, primed word-stem-completion items can be completed with several solutions, thus allowing individual cognitive processes to vary considerably.[4] They estimated reliability of different implicit and explicit measures in several experiments and found patterns that generally supported this hypothesis. However, they argued that not all implicit measures have low reliability. Of particular importance, they found that priming measures that imposed at least moderate processing constraints showed reliability equivalent to that of explicit measures. These included priming tasks involving word identification at brief exposures (17–67 msec) and degraded picture identification. They also noted that measures requiring speeded responding in tasks with specific performance goals should also have acceptable reliability. So findings of poor reliability may be specific to the type of priming measure used, and individual differences may be measurable for some priming measures.

Consistent with the conclusion of Buchner and Wippich (2000), my own research has demonstrated acceptable reliability in repetition priming measures (Woltz, 1988, 1990a, 1990b, 1999; Woltz & Shute, 1993). Most of this evidence pertains to repetition priming in a word-meaning comparison

[3]There is no universal agreement that repetition priming and procedural skill acquisition represent the same underlying processes (Kirsner & Speelman, 1996; Schwartz & Hashtroudi, 1991).

[4]The word-stem-completion task presents people with the first three letters of a word, followed by blank dashes for the remaining letters (c o n _ _ _ _). Items can usually be completed by more than one word (e.g., content, conduct, consent …), but only one solution has been primed by previous exposure to the word.

task. On each trial, participants are presented with two words and instructed to press the *L* key if the words are alike in meaning and the *D* key if they are different. Priming is assessed by facilitation in response latency and accuracy in identically repeated trials, relative to nonrepeated trials. In these studies, repeated trials are lagged anywhere from a few seconds to many days. Of particular interest, facilitation from a single repetition was statistically significant at lag intervals up to 28 days, even though participants had little or no explicit recollection of specific trial content (Woltz & Madsen, 2000; Woltz & Shute, 1995). This suggests that implicit memory changes that facilitate identically repeated events of this kind are remarkably persistent (see also Sloman, Hayman, Ohta, Law, & Tulving, 1988).

The performance goals of this priming task are relatively constrained, which, according to Buchner and Wippich (2000), is a critical feature for reliability in implicit memory measures. In addition, this task measures priming with respect to relatively complex conceptual processes compared to the simpler perceptual processes central to many priming tasks. Reliability was estimated with this priming task in several studies (Woltz, 1988, 1990a, 1990b, 1999; Woltz & Shute, 1993). The reliability estimates were generally not as high as for working memory and explicit memory tasks, but they were high enough to suggest measurable individual differences (internal consistency $.60 < r_{xx'} < .87$ under various trial conditions and repetition lags).

Repetition Priming Differences in Relation to Skill-Learning Differences

Given that moderately reliable individual differences exist in some forms of repetition priming, the important question is whether these differences represent aptitude for more complex forms of learning. Several previous investigations have suggested an affirmative answer to this question.

To investigate the hypothesis that repetition priming differences might further explain skill-acquisition differences subsequent to the initial declarative learning stage, Woltz (1988) estimated the relation between priming on the word comparison priming task and performance on a sequential rule-application skill. Consistent with prior evidence about general cognitive abilities and early skill performance, working memory performance predicted early skill-acquisition differences. In contrast, repetition priming differences increasingly predicted skill performance with practice.

This finding was replicated and extended in a subsequent study (see Woltz, 1999). The same relations were investigated using additional repetition priming and skill-acquisition tasks from the verbal-, numeric-, and spatial-content domains. In addition, explicit memory performance (recognition of prior task events) was measured within the priming task. A latent factor representing the priming measures had statistically significant relations with skill learning, but the explicit recognition factor did not. Also, as found previously (Woltz, 1988), priming differences increasingly predicted skill performance over practice, and this was opposite the trend of general cognitive ability measures.

These studies and others (e.g., Woltz & Shute, 1993) have suggested that repetition priming differences are associated with some forms of learning that entail repetitive practice. Given current theory regarding memory systems underlying repetition priming and procedural skill acquisition, this association could reflect common underlying learning mechanisms.

PRIMING PROCESSES, WORKING MEMORY, AND LANGUAGE COMPREHENSION

Some of the most prominent contributions in recent literature on individual differences pertain to the relation between working memory and language comprehension. Early demonstrations of the relation between these constructs were published by Daneman and Carpenter (1980, 1983). They developed the working-memory-span measure based generally on Baddeley and Hitch's (1974) model of working memory, and they investigated its relation to language comprehension. Their findings suggested relatively high correlations between working memory span and various measures of reading comprehension. These findings in part led to numerous investigations of individual differences in working memory in relation to cognitive development and aging and to other forms of learning and problem solving.

Although the initial estimates of the relation between working memory span and reading averaged close to $r = .7$ (with some as high as .9), a meta-analysis of all the correlation estimates published up to 1995 revealed slightly lower estimates (Daneman & Merikle, 1996). The average weighted Pearson product–moment correlation for verbal working-memory-span measures was .41 for predicting global comprehension measures and .52 for predicting specific comprehension measures.[5] Two additional findings from this meta-analysis clarified some issues regarding the aptitude processes underlying these relations. First, working memory measures that demanded concurrent storage and processing were more highly related to language comprehension than were measures that demanded storage alone (i.e., short-term memory measures such as digit or word span). Second, working-memory-span measures that involved numeric–mathematical process and storage were almost as predictive of

[5] Global comprehension tests were mostly standardized measures of reading and vocabulary, such as the Verbal Scholastic Achievement Test and the Nelson–Denny Reading Test. Specific comprehension tests were mostly experimental measures of integration processes, such as pronoun reference comprehension, ambiguity resolution, and inference making.

specific language comprehension measures (r = .48) as were working-memory-span measures that involved words and language processes (r = .52). These findings are consistent with the conclusion that working memory represents a general (not content-specific) cognitive workspace for concurrent processing and storage demands that appears to be involved in language comprehension as well as other complex learning activities.

Old and New Models of Working Memory

Most of the individual differences research on working memory has been based on the construct definition introduced by Baddeley and Hitch (1974; also see Baddeley, 1986). The model proposed by Baddeley and Hitch includes two storage systems, one for verbal information and one for visuospatial information, and a central executive system that serves to coordinate the storage systems and processes that operate on information held by them. Although Baddeley described a passive element to working memory, the defining feature of the three primary working memory components seems to be their dependence on attention (Baddeley, 1993).

Despite the popularity of Baddeley's (1986) definition of working memory, which focuses on limited-capacity attention processes, some theorists have noted its limitation for explaining certain complex forms of learning and performance. The common theme of these criticisms and corresponding alternative proposals is that attention-driven storage and processing models of working memory cannot "hold" the amount of information that people seem to have readily available in complex cognitive activities.[6] Notable examples of alternative working memory definitions to counter this perceived limitation include the ACT theories of John Anderson (1983, 1993), recent work of Ericsson and Kintsch (1995; Kintsch, Patel, & Ericsson, 1999), and a working memory model proposed by Cowan (1999).

Although there are important differences among these alternative conceptualizations of working memory, they share several important features. First, working memory is defined either wholly or in part with respect to processes that operate on existing long-term memory structures (e.g., Anderson, 1983, defined working memory as the currently active elements of long-term memory). Second, the effective capacity of working memory is seen to be large compared to that of earlier theory. Related to this, unlike the conventional notions of short-term and working memory, these new conceptualizations reject the notion of all-or-nothing storage. Instead, information is available in working memory on a gradient, with some information being more available than others. Finally, in addition to attention-driven processing, some forms of implicit or nonattentional processing appear to be central to these models (e.g., spreading activation).

Given the importance of the working memory construct in current cognitive theory, and given recent trends toward defining working memory limits in part by activation or similar processes, it seems important to consider such processes in terms of their potential as aptitude constructs. However, methods of measuring individual differences in activation processes in working memory have not been well developed compared to methods for studying attention-driven processes of working memory. As will be discussed next, semantic priming methods from experimental research appear to have the most promise, but relatively little evidence exists about individual differences in such tasks.

Individual Differences in Semantic Priming and Language Comprehension

Although there is relatively little existing evidence regarding individual differences in semantic or associative memory activation, there is evidence from several decades of experimental research that addresses various questions about the existence and nature of activation and related processes (e.g., Anderson & Reder, 1999; McNamara, 1992; Neely, 1991). What little evidence there is on individual differences suggests that they may represent an important and somewhat unique aptitude construct.

One of the primary methods of investigating memory activation in experimental research has relied on semantic priming tasks.[7] Semantic priming tasks measure performance facilitation from prior semantically related processing. An early and often-cited example of this is that words such as *butter* are read and pronounced faster when preceded by related words such as *bread* (Meyer & Schvaneveldt, 1976). Individual differences measured by a semantic priming task are considered here.

Woltz (1990b) investigated semantic priming in the word-meaning comparison task described earlier in the dis-

[6]Baddeley has always acknowledged the role of long-term memory in allowing information outside of working memory to be readily available. In addition, he has described passive storage of information in working memory. Thus, these criticisms are to some extent handled by Baddeley's model. Nevertheless, the emphasis in Baddeley's model on attention-driven storage and capacity limits has led some to propose alternatives that include more long-term memory functions, which directly impact immediate processing capabilities attributed to working memory.

[7]Fan tasks have also been used to investigate memory activation. This paradigm tests the notion that verification time for any single fact in memory would be a positive function of how many unrelated facts are known about the subject (Anderson, 1983). The finding that longer verification times occurred for facts with more associates supported the conclusion that a finite amount of activation is divided among all distinct associative links for memory nodes (Anderson, 1983). Both semantic priming and fan tasks have been used in individual differences research, although the fan task evidence proved to more controversial (Cantor & Engle, 1993; Conway & Engle, 1994).

cussion of repetition priming. When a word comparison trial such as *moist damp* was preceded within a few trials by a semantically related comparison (e.g., *soggy wet*), response latency facilitation averaged about 90 msec. Internal consistency reliability for this priming effect was $r_{xx'} = .69$, so individual differences were evident. Of interest here, these individual differences were not highly correlated with working-memory-task performance (r = .27 and r = .32, respectively, with two working memory measures designed to represent Baddeley's, 1986, working memory model).

Subsequent research investigated this form of semantic priming as an aptitude for text comprehension in sixth-grade students (Larkin, Woltz, Reynolds, & Clark, 1996). This study estimated the relation between both repetition and semantic priming and a global measure of reading ability (Stanford Achievement Test[8] reading scores). In addition, explicit memory performance was assessed (episodic recall for prior processing events that produced the priming). The average facilitation in response latency for positive-match trials was over 100 msec for the 60 sixth graders tested. Priming was assessed on two different occasions, so the reliability estimate of .62 included measurement error associated with both items and occasions.

The semantic priming measure had a higher correlation with Stanford Achievement Test Reading (r = .51) than it did with Stanford Achievement Test Math (r = .32), a finding consistent with the idea that spreading activation in semantic memory is an aptitude process particularly relevant to language comprehension. The semantic priming measure also predicted teacher ratings of the students' reading ability (r = .49). In contrast, the repetition priming measure did not correlate with any of the criteria.

A final result of interest in this research was that the explicit memory measure also predicted Stanford Achievement Test Reading (r = .45), Stanford Achievement Test Math (r = .47), and teacher ratings of reading ability (r = .51). Furthermore, semantic priming and recognition had only a modest relation (r = .33), and both made statistically significant contributions in a multiple-regression analysis predicting reading achievement. In total, these findings suggest that individual differences in semantic activation have a substantial relation with global reading achievement and that this relation is somewhat independent of that between explicit memory processes and reading.

Larkin (1996) conducted further research on this issue by contrasting the relation of semantic priming and reading with that of conventional working memory and reading. In contrast to the first study (Larkin et al., 1996), she measured specific reading comprehension processes, such as disambiguation (Daneman & Carpenter, 1983) and anaphoric reference comprehension (Gernsbacher, 1989). In a sample of 94 sixth graders, the semantic priming measure had a higher correlation with the reading comprehension composite (r = .61) than did the working memory measure (r = .44). The correlation between semantic priming and working memory measures was modest (r = .25), and both made statistically significant contributions in a multiple-regression analysis predicting reading comprehension from measures of semantic priming, working memory, vocabulary knowledge, and processing speed.

These findings are consistent with the view that implicit activation processes in long-term memory play a role in language comprehension that is distinct from that of attention-driven working memory processes. This view corresponds to some recently proposed models of working memory that contend attention and activation processes impose different constraints on cognition (e.g., Cowan, 1999). Clearly, more empirical tests of these ideas in the form of aptitude–learning relations are needed. Nevertheless, the evidence that does exist suggests that temporary memory activation represents a potentially important aptitude process in which learners differ, one that corresponds to current cognitive theory about processing-capacity limits, and one that may be relatively distinct from conventional aptitude constructs.

IMPLICATIONS FOR FUTURE RESEARCH

About 10 years ago in the *Educational Psychologist*, Richard Snow described the necessary constituents of an aptitude theory (Snow, 1992). He saw aptitude theory as providing a valuable link for fields of psychology and education between the understanding of individual characteristics and environmental and instructional contexts. It is informative to use this comprehensive framework to consider both the status of the implicit cognitive processing constructs discussed here and the needs of future research in this area. Snow (1992, 1998) described 12 constituents of an aptitude theory. The first 7 pertain to theoretical issues of aptitude constructs, and the last 5 pertain to practical issues of measurement, methodology, and decision rules for application. Only the first 7 are considered here.

The first two of Snow's (1998) constituents of aptitude theory correspond to traditional concepts of convergent, discriminant, and predictive validity relations. Regarding these, there exists initial evidence of predictive relations for two distinct forms of priming (repetition and semantic) with complex forms of learning (skill acquisition and language comprehension). More evidence is needed regarding these relations, as well as other relations that stem from theoretical descriptions of various learning processes. Equally as important, future research aimed at developing aptitude constructs in this area must estimate relations with conventional aptitude constructs. Again, both convergent and discriminant relations investigated should correspond to predictions based on cognitive and aptitude theory.

Snow (1998) described the next constituent of aptitude theory, process explanations, as perhaps the most important

[8]Stanford Achievement Test is published by Psychological Corporation (1990). Scores were obtained from fifth-grade testing.

issue underlying theory-oriented aptitude research. Given the extensive body of experimental memory research defining implicit learning and memory mechanisms, process explanations of the constructs discussed here are relatively well developed. However, these process explanations are at a microlevel. Snow's concept of *process explanation* for aptitude constructs was broader, including both micro- and macrolevel descriptions. Consistent with this broader view, future research in this area needs to describe at a macrolevel how implicit learning mechanisms function in more complex instructional settings, in interaction with other processes (including conative and affective processes).

Another set of related constituents in Snow's (1998) framework of aptitude theory represents the understanding of short-term malleability of aptitudes by various interventions and long-term developmental changes in aptitudes. In conjunction with other validity evidence, these issues help to define the fundamental nature of the constructs in question. Some evidence exists regarding developmental trends in implicit learning and memory phenomena, but relatively little evidence exists to address the short-term malleability of these processes in individuals. This may be a particularly important issue with respect to this category of constructs. Whereas many general intellectual aptitudes tend to show definitive developmental patterns and be difficult to affect with short-term intervention strategies, implicit processes may be quite different in these respects. These represent important topics for future research.

The final set of constituents of Snow's (1998) framework refers to differential predictive evidence, such as aptitude–treatment interaction (ATI), and other evidence that establishes boundary conditions of the aptitude constructs (e.g., understanding how aptitude relations with learning may be situated in specific contexts). To my knowledge, no evidence of this type exists for implicit cognitive processes. Clearly, these constituents represent an important aspect of construct validity to be addressed in future research. In characterizing the importance of ATI evidence for aptitude theory, Snow (1998) noted, "To demonstrate that the relation of an ability measure to other measures can be manipulated experimentally is to show that the ability construct is understood" (p. 97).

SUMMARY AND CONCLUSION

In his last 2 decades, Richard Snow's contributions seemed to be focused more on the understanding of aptitude complexes that included conative and affective aptitudes, rather than the further search for new or revamped cognitive aptitudes. However, he continued to advocate the importance of advancing process descriptions of all aptitude domains.

Implicit learning and memory processes represent an intriguing class of cognitive mechanisms that have received relatively little attention in aptitude theory compared to attention-related and general intellectual ability constructs. As noted earlier, this could reflect the fact that implicit cognitive processes by nature are relatively inaccessible to awareness and introspection by both learners and learning theorists. The fact they we have relatively little detailed understanding of the roles of implicit learning processes in educational settings may be more a function of the difficulty of thinking about these processes rather than their importance. Questions of whether implicit cognitive processes represent meaningful aptitudes for learning can only be answered by adapting experimental methods from laboratory research to the study of individual differences in instructional contexts.

Although there is relatively little evidence about individual differences in implicit cognitive processes, most of the existing evidence suggests that individuals differ reliably in these processes and that these differences are related to some forms of complex learning. Individual differences in repetition priming are predictive of middle and later stages of cognitive skill acquisition in several content domains, and semantic priming differences are predictive of various aspects of reading comprehension. These predictive relations appear to be largely independent of those involving other aptitude constructs. In total, the evidence suggests that learner differences in implicit cognitive processes might be an important area for future aptitude and learning research.

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